```
#include <stdio.h>
#include <conio.h>
#include <stdlib.h>

int i, j, k, a, b, u, v, n, ne = 1;
int min, mincost = 0, cost[9][9], parent[9];
int find(int);
int uni(int, int);
void main() {
  printf("\n\tImplementation of Kruskal's Algorithm\n");
  printf("\nEnter the no. of vertices:");
  scanf("%d", & n);
  printf("\nEnter the cost adjacency matrix:\n");
  for (i = 1; i <= n; i++) {
    for (j = 1; j <= n; j++) {
      scanf("%d", & cost[i][j]);
      if (cost[i][j] == 0)
        cost[i][j] = 999;
    }
  }
  printf("The edges of Minimum Cost Spanning Tree are\n");
  while (ne < n) { //edges < vertex - 1
    for (i = 1, min = 999; i <= n; i++) {
      for (j = 1; j <= n; j++) {
        if (cost[i][j] < min) {
          min = cost[i][j];
          a = u = i;
          b = v = j;
        }
      }
    }
    u = find(u);
    v = find(v);
    if (uni(u, v)) {
      printf("%d edge (%d,%d) =%d\n", ne++, a, b, min);
      mincost += min;
    }
    cost[a][b] = cost[b][a] = 999;
  }
  printf("\n\tMinimum cost = %d\n", mincost);
  getch();
}
int find(int i) {
```

```
  while (parent[i])
    i = parent[i];
  return i;
}
int uni(int i, int j) {
  if (i != j) {
    parent[j] = i;
    return 1;
  }
  return 0;
}
```